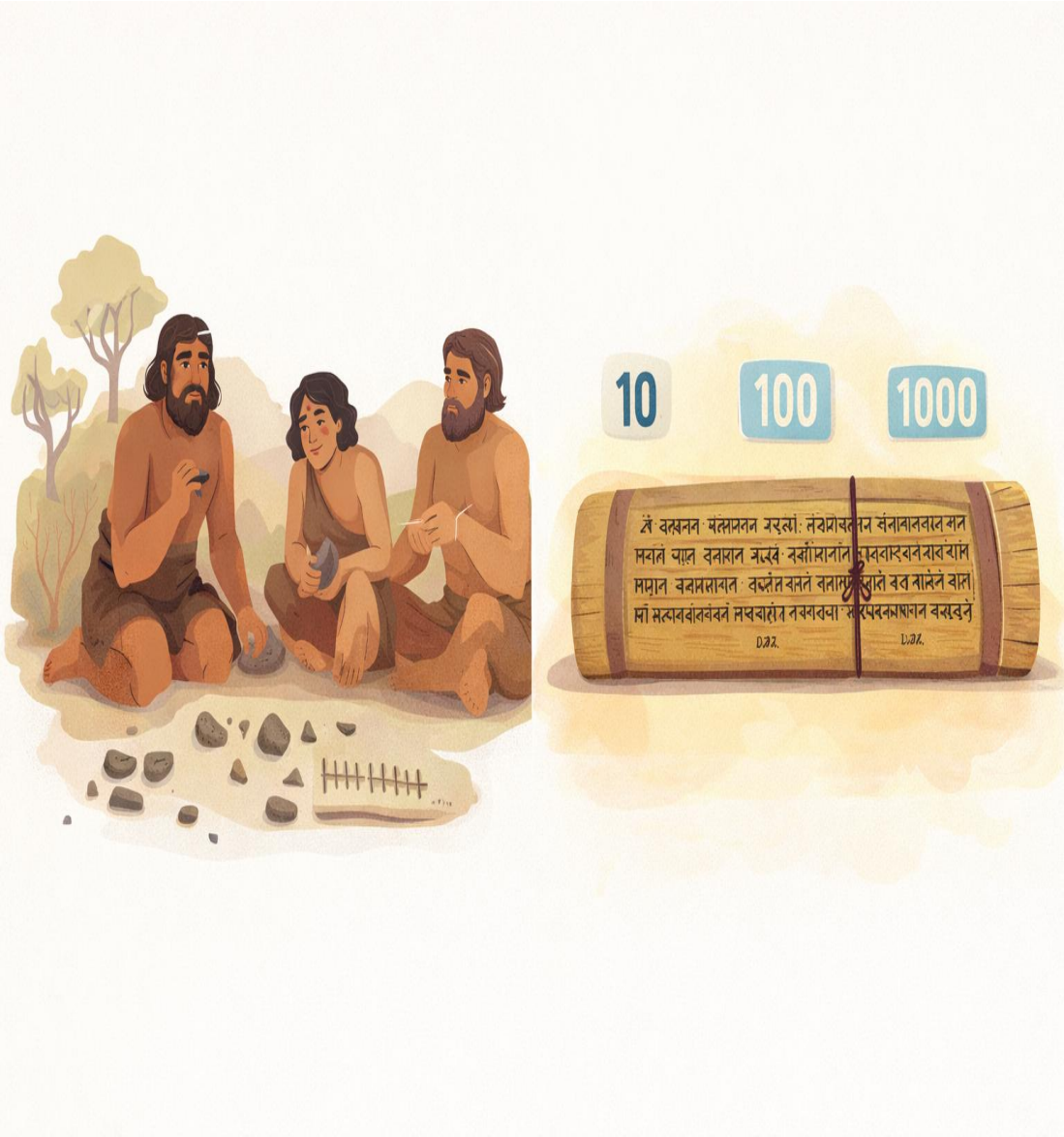
Chapter 4: Decimal Numbers

Background -Origins and Historical Spread

- Understanding numbers was central to early civilizations.
- Surprisingly, India has no early independent text solely on arithmetic, unlike geometry (Śulbasūtra).
- Counting and arithmetic require:

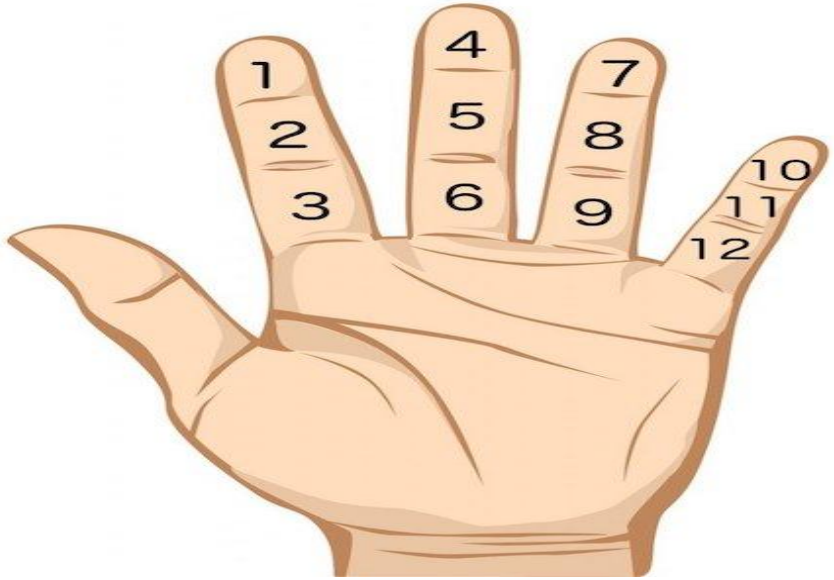
1. Logical thinking
2. Abstraction
3. Concept of cardinality

- The decimal number system (base 10) originated in India well before the final compilation of the Rgveda, specifically before the 12th–11th century BCE.



Background -Early Numeracy in India

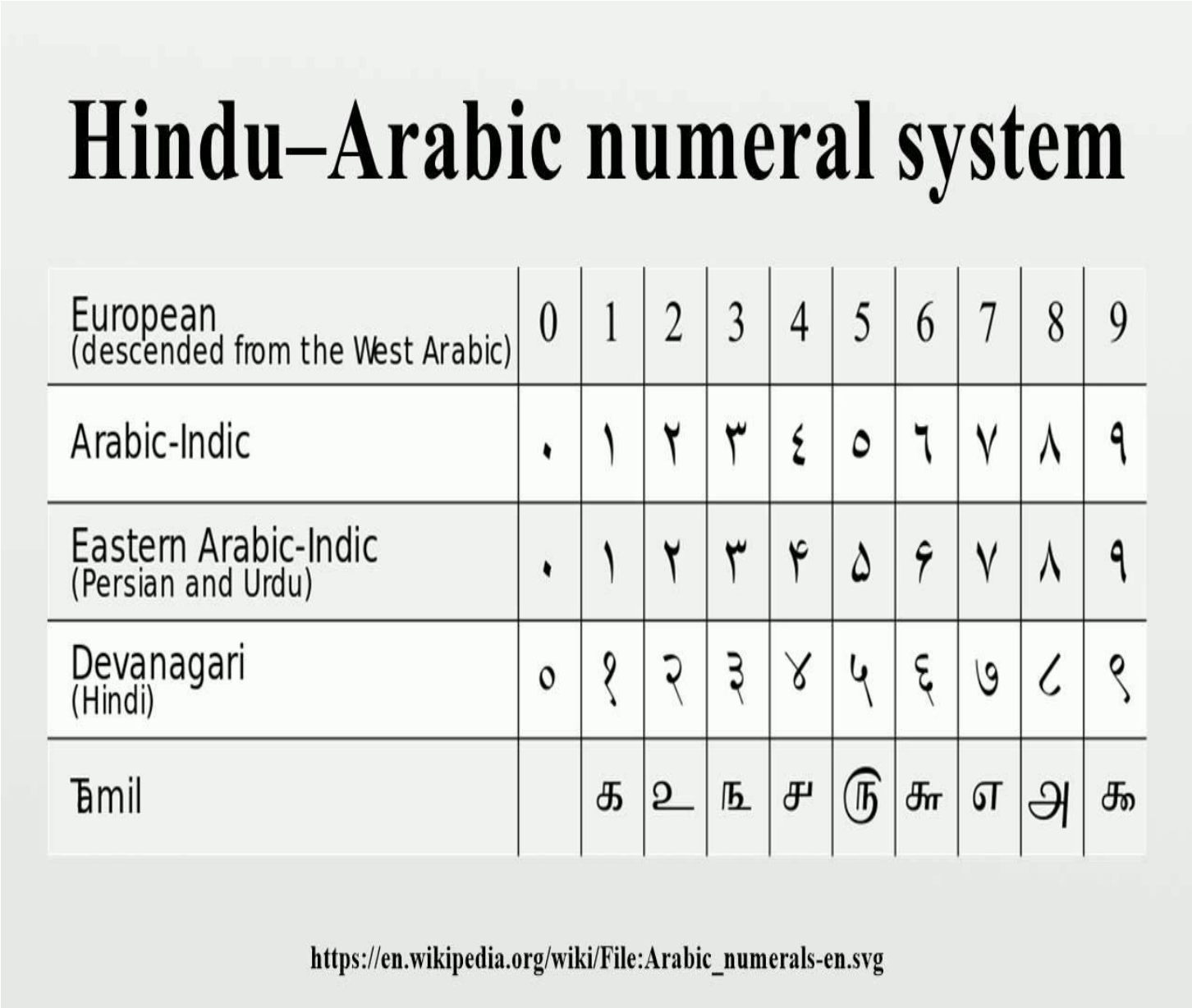
- Early Indians did not write a separate book only about numbers.
- Counting likely developed naturally in daily life before formal mathematics.
- Arithmetic became a basic intellectual skill requiring abstraction and logic.



Background -Birth of the Decimal (Base-10) System in India

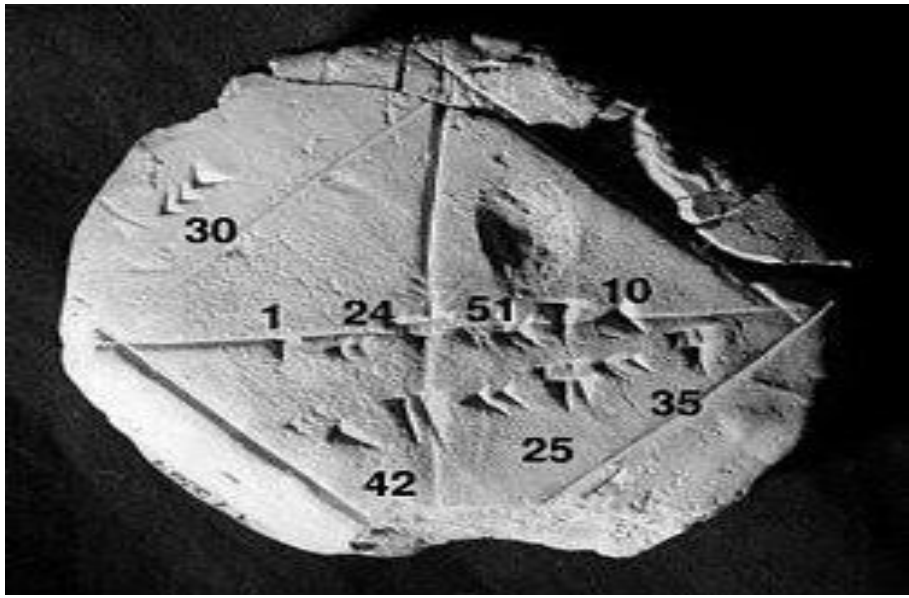
- The **decimal (base-10) system** originated in India before 1200–1100 BCE.
- The *Rigveda* provides early evidence of systematic decimal naming.
- Numbers were clearly organized in powers of 10 (10, 100, 1000…).
- Base-10 remained central in Indian mathematics throughout history.

Example of place value:
345 = (3 × 100) + (4 × 10) + (5 × 1)



Background -Babylonian Base-60 System

- Around 1800 BCE, Babylonians used **base-60 (sexagesimal)** notation.
- They introduced positional notation for powers of 60.
- Unlike India, fractions remained sexagesimal in Mesopotamia.
- **Example:**
  1 hour = 60 minutes
  1 minute = 60 seconds
  360 degrees in a circle (6 × 60)
- For numbers **1 to 9**, Babylonians repeated the symbol for 1.
  Example:
  3 = 1 1 1(three unit symbols)



- For numbers ≥ 60, they used place value.
- Babylonials used base 60:

  Example:Represent 125

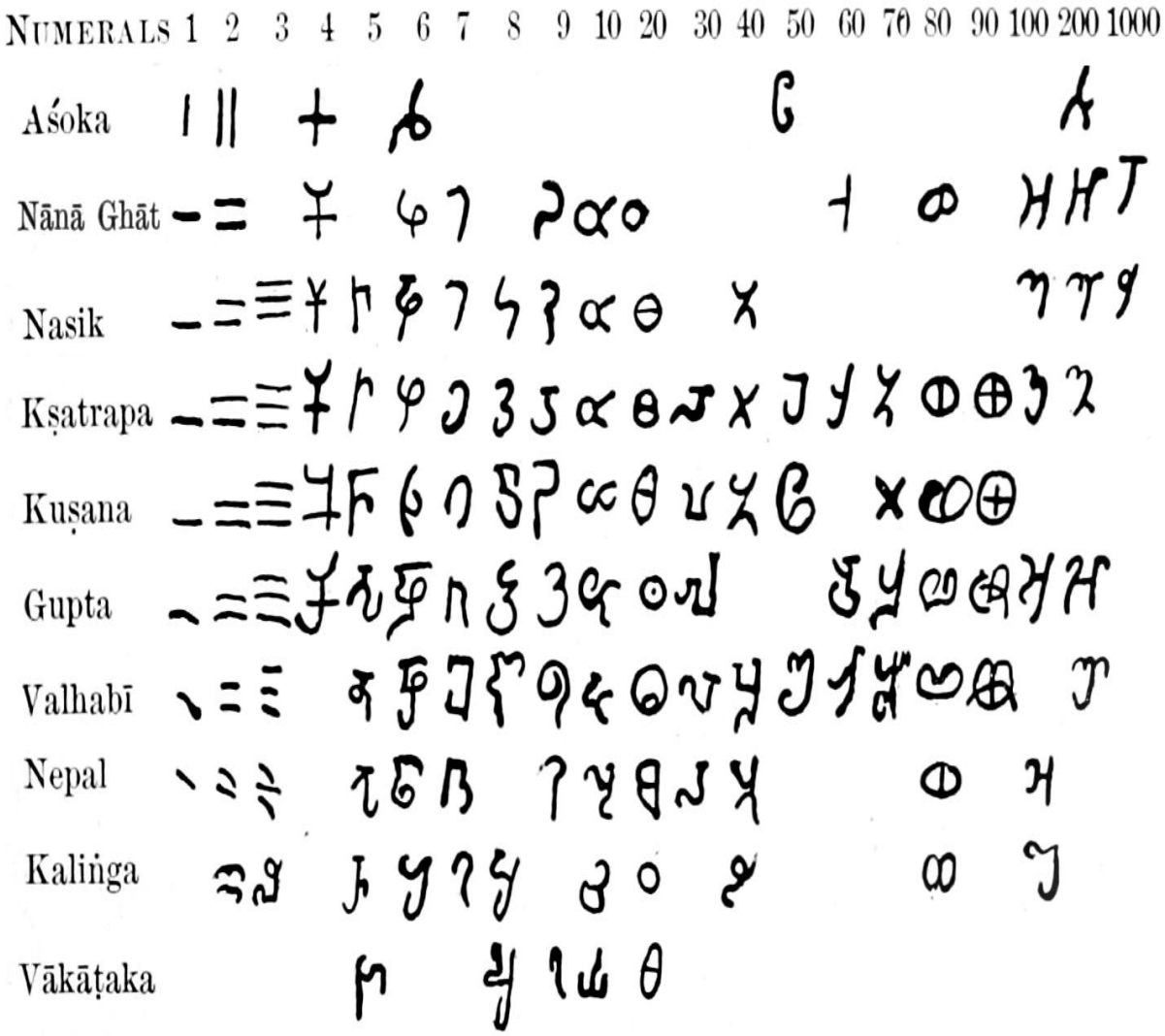
$$125 = 2 \times 60 + 5$$

Background -Indus Valley Number Evidence

- Indus people used literal stroke marks for small numbers (up to 7 or 8).
- Possible use of base-8, but no clear proof.
- Clear decimal structure appears later in Vedic culture.
- **Example of stroke system:**
  ||||| = 5
  Becomes difficult for large numbers like 78.

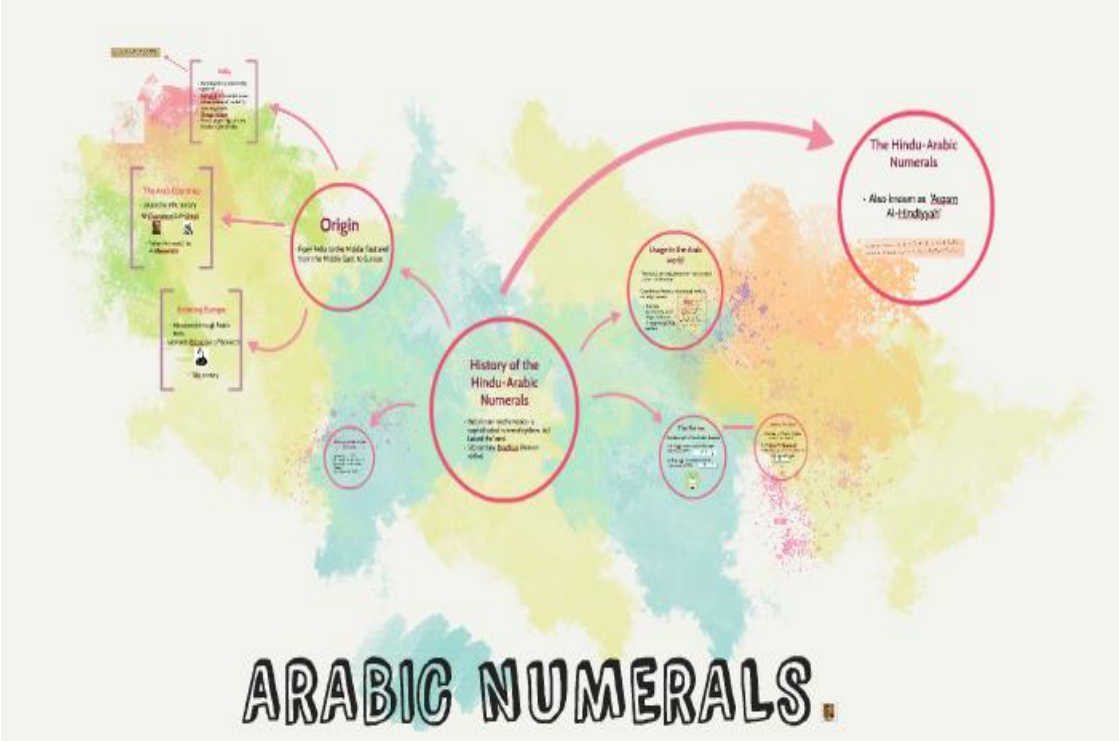
Background -Development of Written Numerals (Brahmi Script)

- After 3rd century BCE, numbers appeared in **Brahmi script** inscriptions.
- Early Brahmi numerals were not fully positional.
- By the 6th century CE, strict positional decimal notation appears clearly.
- **Example:**
  205 means
  2 hundreds + 0 tens + 5 ones
  Zero became very important.



Background –Indian Mathematical Foundation

- Brahmagupta (7th century) established essential rules for arithmetic, extending them to **including zero and negative numbers.**
- Bhaskara II (12th century) furthered these algebraic treatments in his treatise Bījagaṇita, though formal "logical rigour" as understood today was a later development of the Nila school.
- The 16th-century text Yuktibhāṣā provided an abstract recapitulation of numbers, defining them through the property of succession (the idea that every number has a unique successor), a concept that mirrors the later Peano axioms of the 19th century



- Spread to Islamic lands, then to Europe.
- Introduced to Europe by Fibonacci (13th century).
- Later called “Arabic numerals.”

Background –Indian Contributions to Algebra & Calculus

- Brahmagupta introduced rules for zero and negative numbers.
- Bhaskara II developed algebra systematically.
- Kerala (Nila) school studied infinity ar series.
- Recognized that decimals behave like polynomials.
- **Example (Rule of zero):**

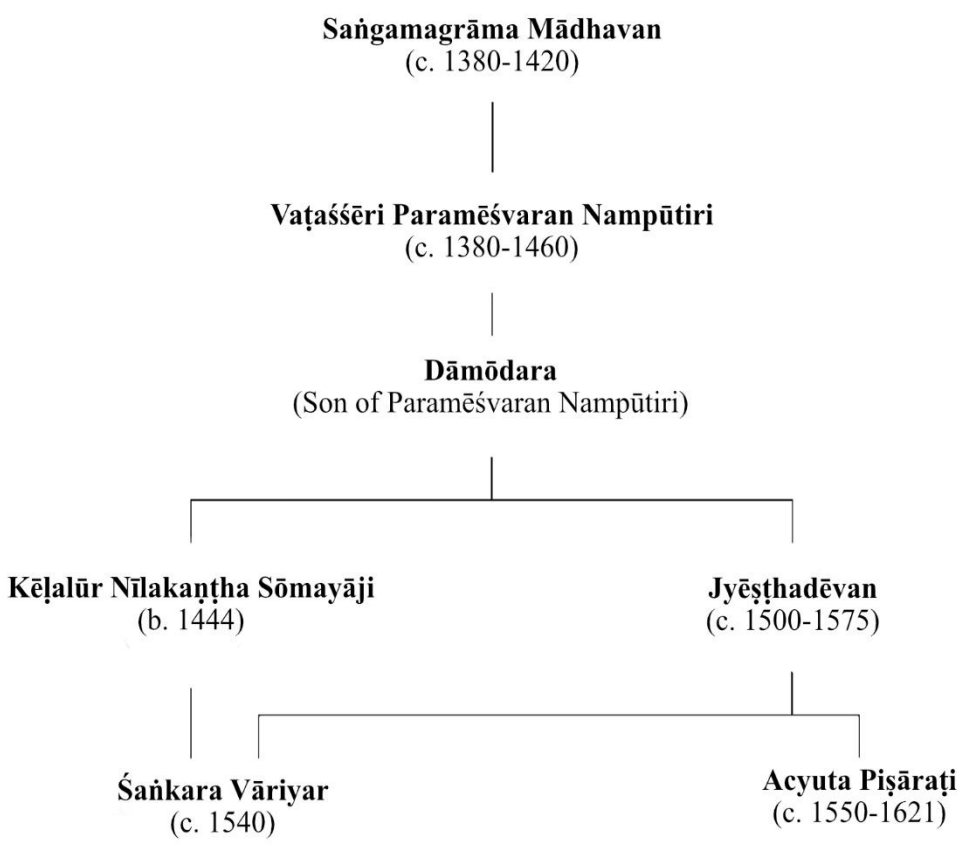
$a + 0 = a$

$a \times 0 = 0$

Example (Infinite series idea)

$1/2 + 1/4 + 1/8 + 1/16 + \ldots \rightarrow$ approaches 1

This later helped in calculus.



Numbers and Based Numbers

- 19th century Europe aimed to give **formal logical foundations** to mathematics.
- Inspired by **Greek logic** and philosophers like **René Descartes**.
- **Giuseppe Peano (1889)** defined axioms for natural numbers.
- Mathematics built from:
  - A **set** (N)
  - Operations (+, ×)
  - Logical rules

Mathematical Example (Peano Idea)

Define a successor function:

$$S(n)=n+1$$

If:

$$S(3)=4$$

Then numbers are built as:

$1, S(1)=2, S(2)=3, S(3)=4, \ldots$

This shows:

- Every number has a **unique successor.**
- Numbers are **infinite.**

Numbers and Based Numbers-Arithmetic Built from Successor

- Addition defined as repeated succession.
- Multiplication defined as repeated addition.
- Division defined using the **division algorithm**.
- **Examples**

Addition (Iterated Successor)

3+2

$S(S(3))=S(4)=5$

Multiplication (Repeated Addition)

$3\times4==3+3+3+3=12$

Numbers and Based Numbers-Decimal Place Value as Polynomial

- Every number in base 10 is a polynomial in 10.
- Coefficients are digits (0–9)
- This is based on repeated division by 10.

Example

Take:5824

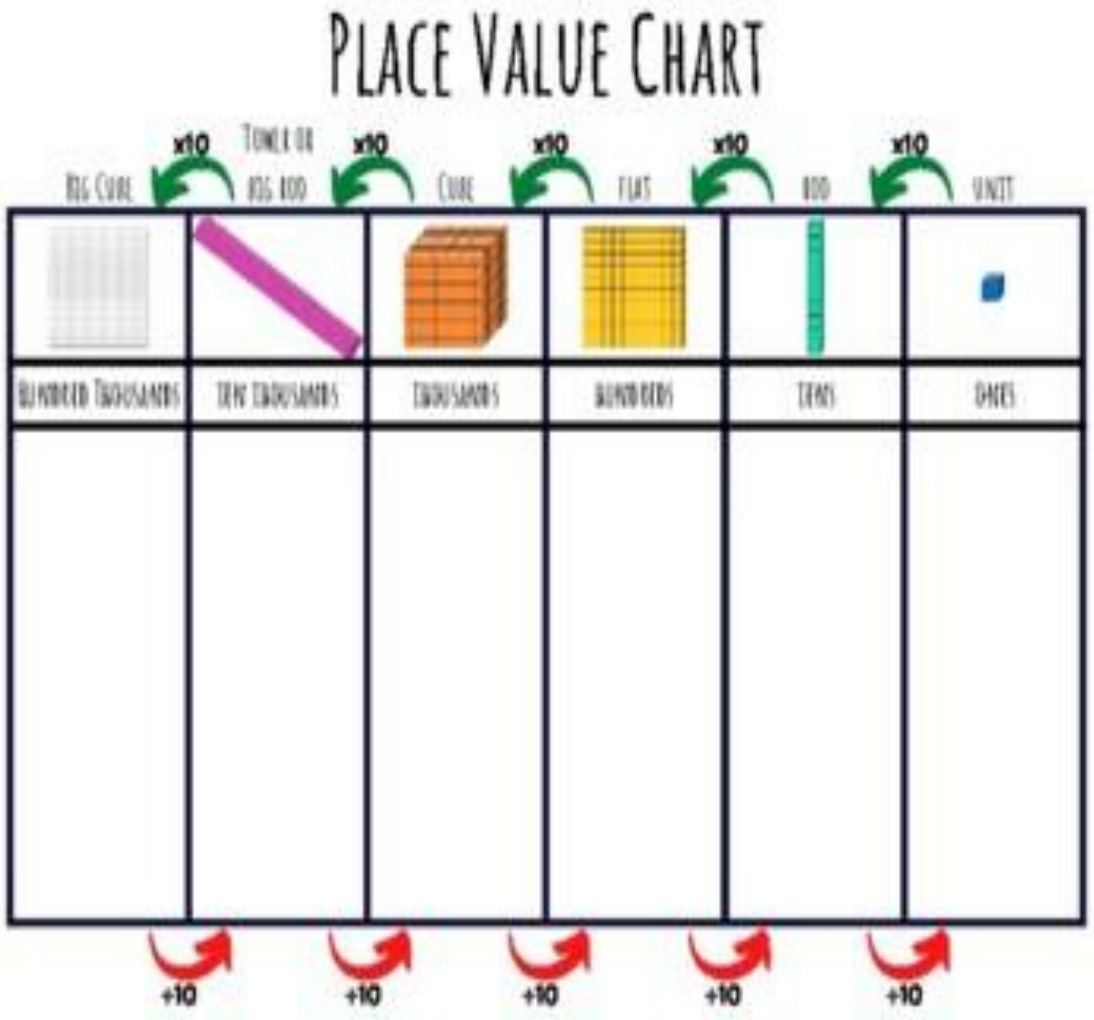
Polynomial form:

$$5824 = 5 \times 10^3 + 8 \times 10^2 + 2 \times 10^1 + 4 \times 10^0$$

So:

$$N = \sum_{k=0}^{n} a_k 10^k$$

This is the foundation of algebra.

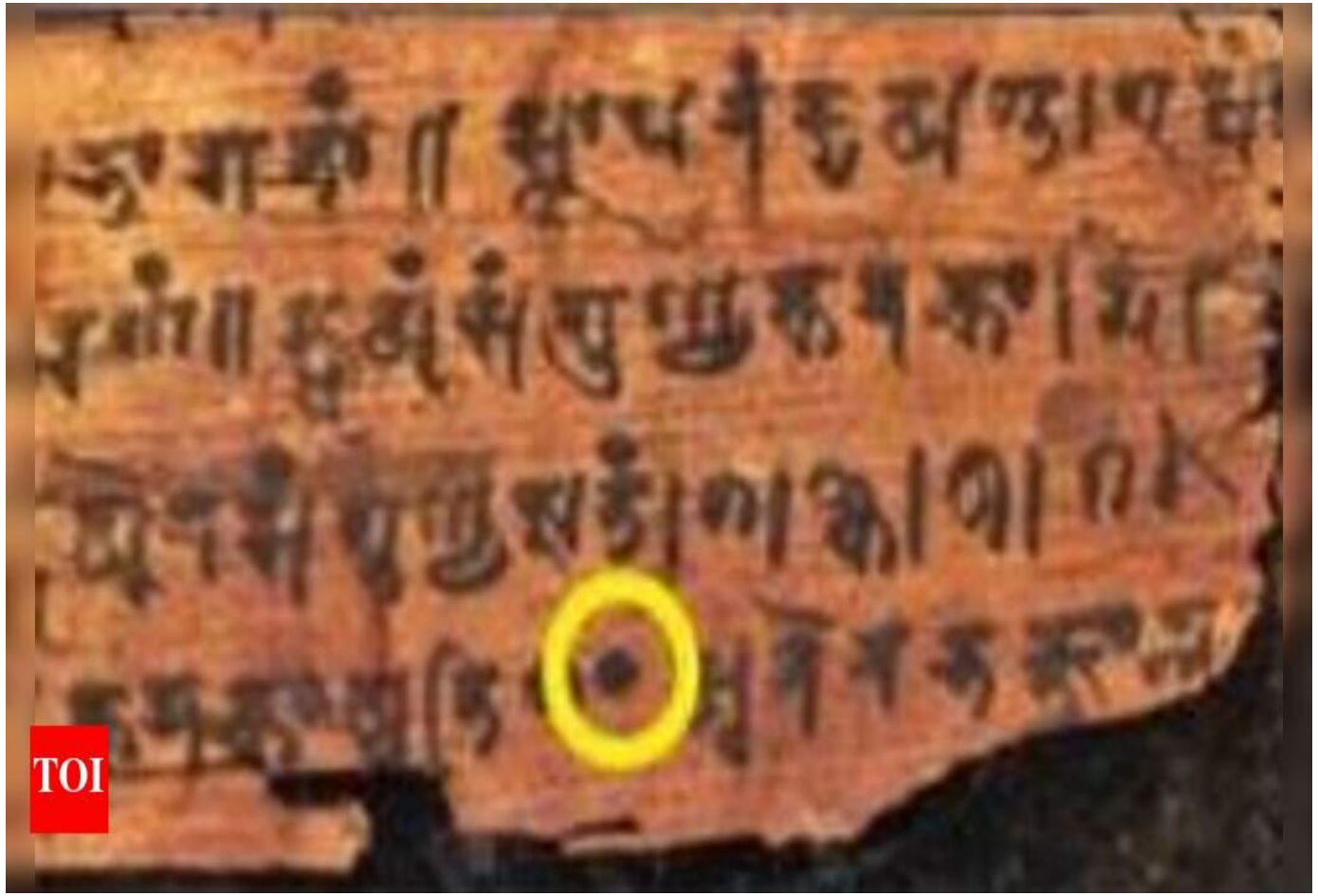


Numbers and Based Numbers-Indian Perspective – Atomic Numbers & Place Value

- In Indian mathematics, numbers **1 to 9** were treated as fundamental or “atomic” numbers.
- All large numbers built using:

Digits 0–9 Place value

- Zero is essential for empty place.
- Example: $1005 = 1 \times 10^3 + 0 \times 10^2 + 0 \times 10^1 + 5$
- Without 0 → confusion.



Numbers and Based Numbers-Yuktibhasha & Logical View of Numbers

- In *Yuktibhasha*, numbers were explained through the idea of **succession** (next number).
- Every number is formed by adding 1 repeatedly.
- Addition is understood as moving forward in counting.
- Subtraction is understood as moving backward.
- This shows a deep logical understanding of arithmetic, not just mechanical rules.

Example:If we start with 9

Forward::9+4=10,11,12,13

Backward:9−4=8,7,6,5

So addition and subtraction come from counting.

Numbers and Based Numbers-Recursive Thinking & Euclidean Algorithm

- A recursive process means repeating the same operation again and again.
- The decimal place-value system itself is recursive.
- Indian mathematicians used recursion in:

1. Division
2. Root extraction
3. Infinite series

- This recursive thinking later supported early calculus in Kerala school mathematics.

Euclidean Algorithm Example

Find GCD(48,18)

48=18(2)+12 18=12(1)+6 12=6(2)+0

So gcd(48,18)=6

This uses repeated division (recursive method).

Peano vs Decimal View

- Numbers built by succession
- Linear growth
- Focus on logic

- Numbers built by place value
- Exponential growth
- Focus on arithmetic

The Place-value Principle and its Realisations-Abstract Idea of Numbers (Beyond Writing)

- Early civilizations did not always write numbers some **spoke them (Vedic India)**, some **wrote symbols (Babylonia)**.

- The decimal representation :N=$(n_k \ldots\ldots n_0)$ is only a **concrete expression** of a deeper abstract idea.

- The abstract structure of numbers:Counting, Arithmetic, Division with remainder.

- Counting and arithmetic evolved together from practical needs like measurement.

The Place-value Principle and its Realisations-Earliest Counting Method One-to-One Matching

- Early counting was done by **matching objects**.
- Example: 5 sheep matched with 5 stones.
- This allowed comparison:Equal, More, Less
- Limitation: Could not easily measure large numbers.
- Solution: Fix a standard base (like 10).

Example: If we fix base 10

To measure 37:$37=3\times10+7$

We compare repeatedly with 10 instead of matching one by one.

The Place-value Principle and its Realisations- The Place-Value Principle (Core Idea)

- Place-value principle:

1. Repeatedly divide by base.
2. Record remainders.
3. Continue until quotient < base.
4. Express number as polynomial in base.

Example:Convert 742 into base-10 polynomial form:

$742 \div 10 = 74$ remainder 2

$74 \div 10 = 7$ remainder 4

$7 \div 10 = 0$ remainder 7

So: $742 = 7 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$

This works for **any base**.

The Place-value Principle and its Realisations-Written Representation – Why It Became Dominant

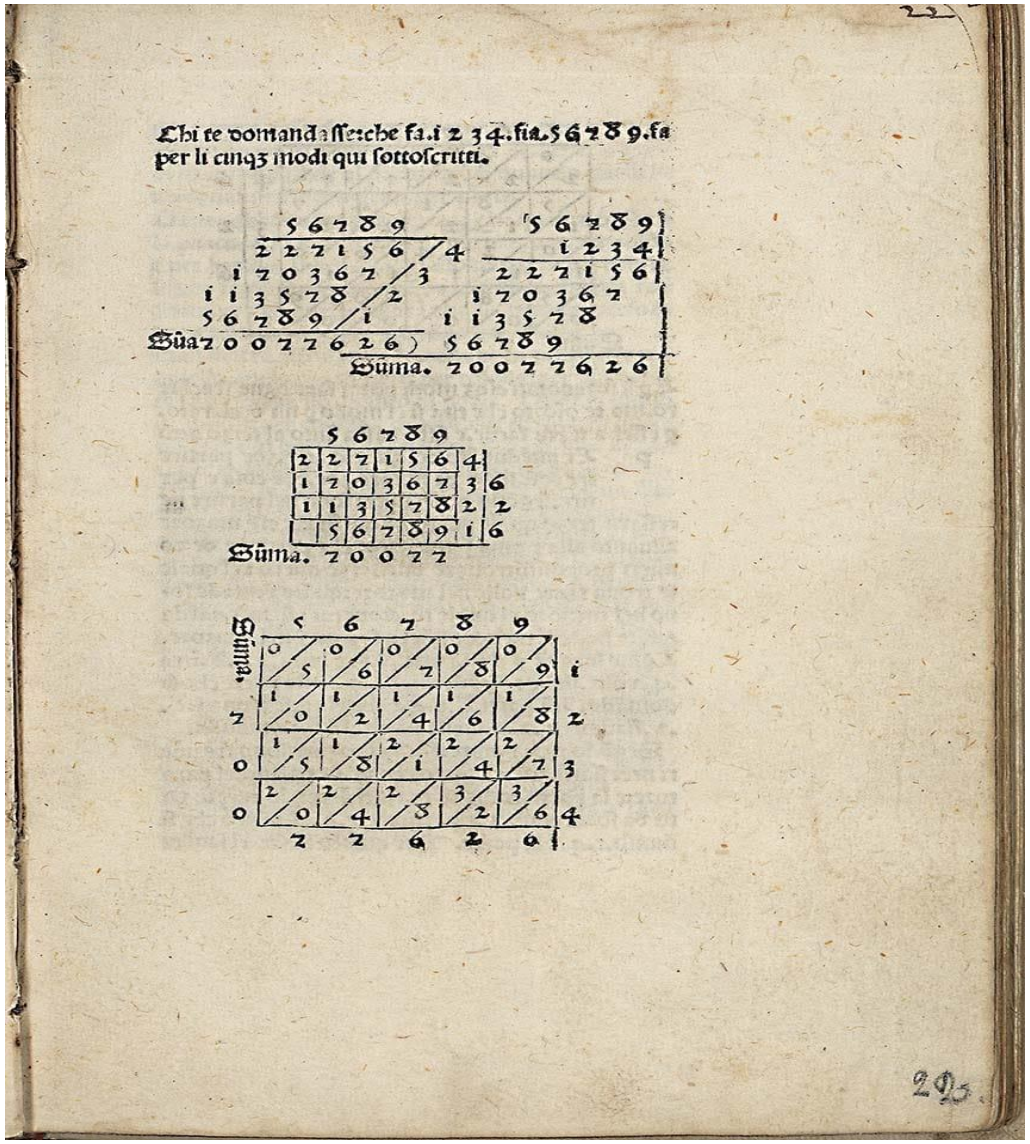
- Advantages of written decimal system:

1. Only 10 symbols (0–9)
2. Very memory efficient
3. Length of number ≈ logarithm of its size
4. No ambiguity
5. Fits perfectly into written text

Example:A 10-digit number

$10^9 < N < 10^{10}$

Instant magnitude estimation.



The Place-value Principle and its Realisations-Oral Number System in Vedic India

In Vedic India:

- Numbers were spoken, not written.
- Atomic numbers had fixed names:

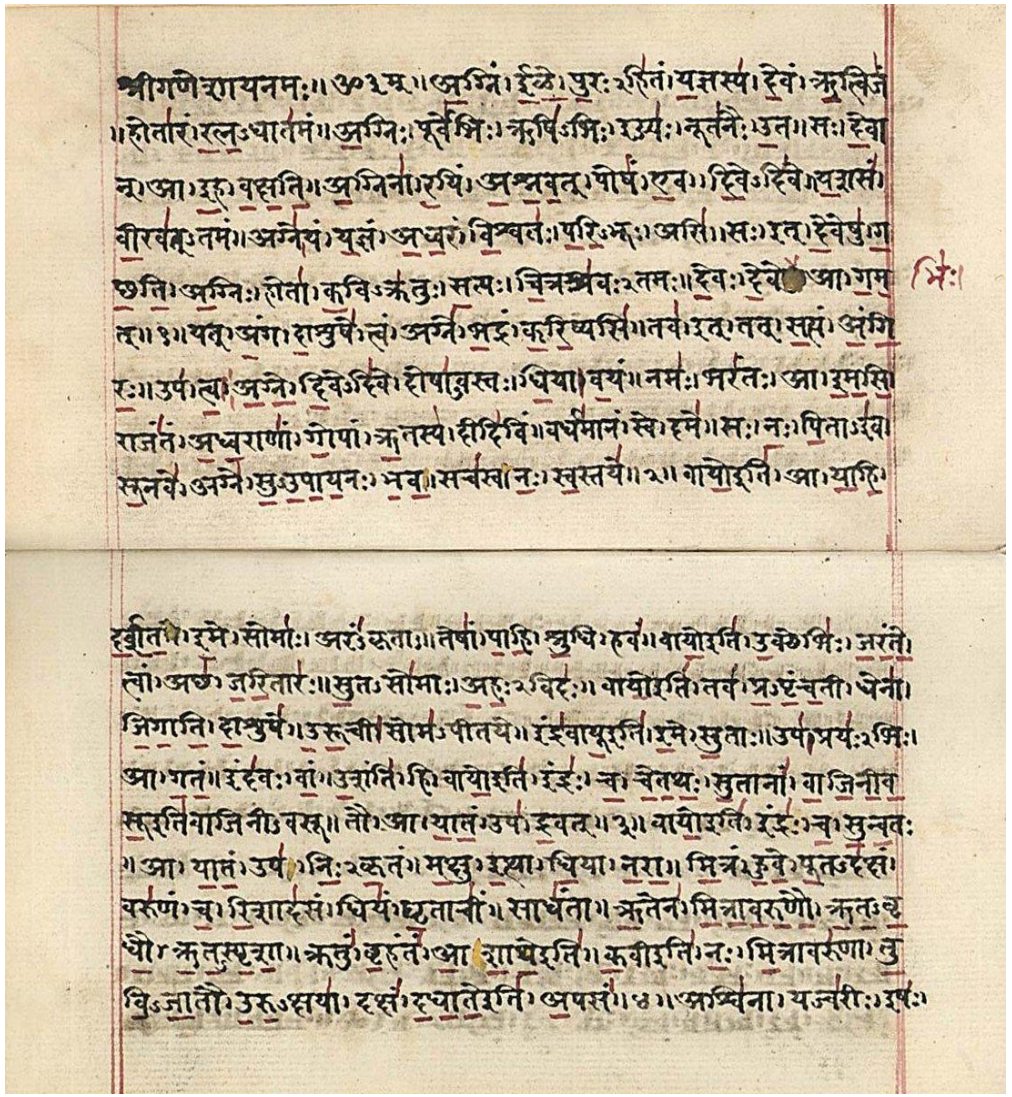
1. eka (1) 2.dvi (2) 3.tri (3)

4.catur (4) 5.pañca (5) 6.ṣaṭ (6)

7.sapta (7) 8.aṣṭa (8)

Rules required:

1. One name → one number
2. Grammar combined words to create large numbers



The Place-value Principle and its Realisations-Naming Powers of 10

To build large numbers:

Step 1: Name atomic numbers

Step 2: Name powers of 10

Step 3: Combine using grammar rules

Example structure:$3\times10^2+5\times10+7$

Becomes linguistic combination meaning:

“Three hundreds, five tens, seven”

This is polynomial expressed in language.

The Place-value Principle and its Realisations-Zero in Oral vs Written System

- Written system requires symbol 0.

- Without 0, place-value collapses.

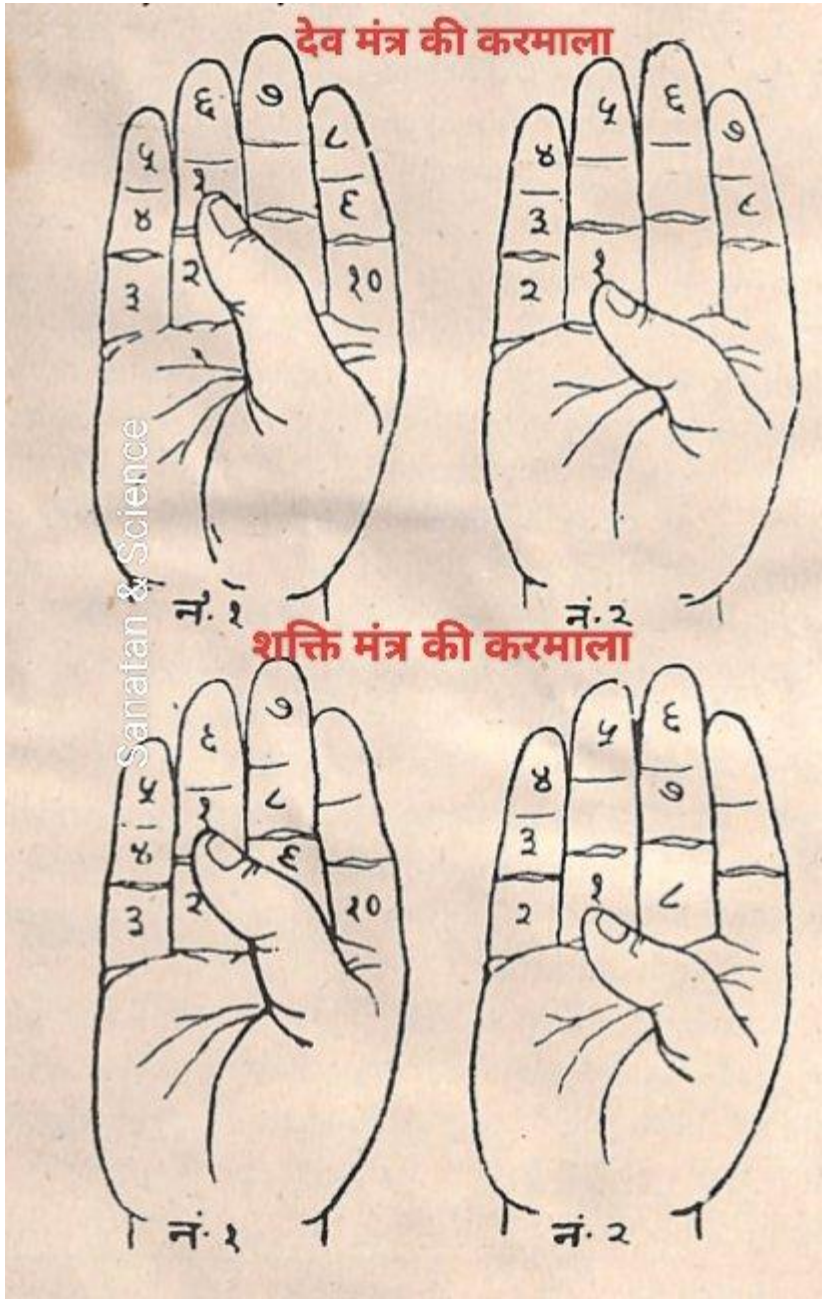
- Example:1005 Needs two zeros.

- But in oral naming:

1. Zero is not needed unless number is zero itself.
2. Spoken numbers correspond directly to polynomial terms.

Other Realisations-Gestural Decimal Counting & Gayatri Mantra (Vedic Tradition)

- One of the **oldest decimal counting methods** in India was used during the recitation of the **Gayatri mantra** from the *Rigveda*.
- The mantra was repeated **108 times (or 1008 times)**.
- Counting was done using a **gestural place-value system**:Right hand → ones place, Left hand → tens place
- When 100 was completed, a blade of grass was moved aside to track hundreds.
- This system:Did not use writing, Did not require spoken numbers, Was independent of language



Other Realisations-Katapayadi Number Systems(c. 7th century CE)

- The Katapayadi system is an old Indian method of writing numbers using letters. It helps in remembering large numbers by turning them into words or verses, and it was useful in astronomy, mathematics, and poetry.
- In this system, certain consonants are given number values from 0 to 9, arranged in four groups starting with ka, ta, pa, and ya. Vowels have no value and are used only to form pronounceable words.
- Example (value of π encoded in a verse):
  भद्राम्बुधसद्धजन्मगणतश्रद्धा स्म यद् भूपगीः
- After converting the consonants into digits:Consonants भ र ब ध स ध ज म ग ण त र ध म य भ प ग
- Digits 4 2 3 9 7 9 8 5 3 5 6 2 9 5 1 4 1 3
- Reading the digits in reverse order gives: π≈3.14159265358979324

Other Realisations-Bhuta-Sankhya System

- The Bhūtasankhyā system is an ancient Indian method of representing numbers using meaningful words related to objects, nature, or philosophy.
- Each word symbolizes a number. Example: Netra (eyes) = 2, Veda (four Vedas) = 4, Graha (nine planets) = 9, Śūnya (void) = 0.
- Mathematicians used these words in Sanskrit verses so that large numbers could be memorized easily in astronomy, poetry, and mathematics.

THE SANSKRIT ALPHABET

The Sanskrit alphabet is organized as follows, reading from left to right:

Vowels (when not combined with consonants)

अ a	आ ā	इ i	ई ī	उ u	ऊ ū
ऋ ṛ	ॠ ṝ	ऌ ḷ			
ए e	ओ o	ऐ ai	औ au		

Consonants (with inherent vowel *a*)

Velar:	क ka	ख kha	ग ga	घ gha	ङ ṅa
Palatals:	च ca	छ cha	ज ja	झ jha	ञ ña
Cerebrals:	ट ṭa	ठ ṭha	ड ḍa	ढ ḍha	ण ṇa
Dentals:	त ta	थ tha	द da	ध dha	न na
Labials:	प pa	फ pha	ब ba	भ bha	म ma
Semivowels:	य ya	र ra	ल la	व va	
Sibilants:	श śa	ष ṣa	स sa		
Aspirate:	ह ha				

(Add-on signs:) ं ṃ (anusvāra) ः ḥ (visarga)

The Choice of a Base

- Human cognition treats **small numbers (below the base)** differently from large numbers.
- Small numbers can be grasped **intuitively at a glance** (e.g., 4 objects).
- Large numbers require **rules, grouping, and place-value structure**.
- Evidence from Indian tradition:Vedic metres: 8–12 syllables per line, Musical tāla cycles: 7–16 beats.
- These suggest that cognition comfortably handles numbers **around 10**.
- Hence, **base 10** is cognitively practical — not just because of 10 fingers, but due to limits of visual and auditory perception.

Example: We can instantly perceive:● ● ● ● → 4 (intuitive perception)

But consider:4739, We cannot "see" 4739 directly. Instead, we process it using place value:$4739=4\times10^3+7\times10^2+3\times10^1+9\times10^0$

Here:Digits 0–9 are atomic (memorised symbols), Large numbers are handled by applying positional rules.

Sample Questions

1. What is meant by the cardinality of a finite set? How is counting related to it?
2. Why does counting require logical thinking and abstraction, even though it appears natural?
3. Why is the absence of an early Indian arithmetic text significant in understanding early mathematical development?
4. What is meant by a fully fledged decimal system?
5. . Explain the place value principle in positional number systems.
6. What is meant by the base of a numeration system?
7. Why is 10 considered a natural base for enumeration in many cultures?
8. What is the difference between a literal number system and a positional number system?
9. Explain the main characteristic of the Babylonian sexagesimal system.
10. What is meant by the realisation of the place value principle?

Sample Mathematical Questions

1. Write the number 4372 using the decimal place value expansion and compare it with the Katapayadi Number System representation.Write the number 125 in base 60.
2. Express 526 using decimal place value expansion and analyze how such numbers could be represented in the Katapayadi system.Show that the decimal number 372 can be written as a polynomial in 10.
3. Represent the number 5 using the tally representation used in early numeral systems.
4. Express the number n in terms of repeated addition.
5. Why does the Babylonian system use base 60 instead of base 10?
6. What are the limitations of tally representation for large numbers?
7. Explain how place value increases the efficiency of number representation.